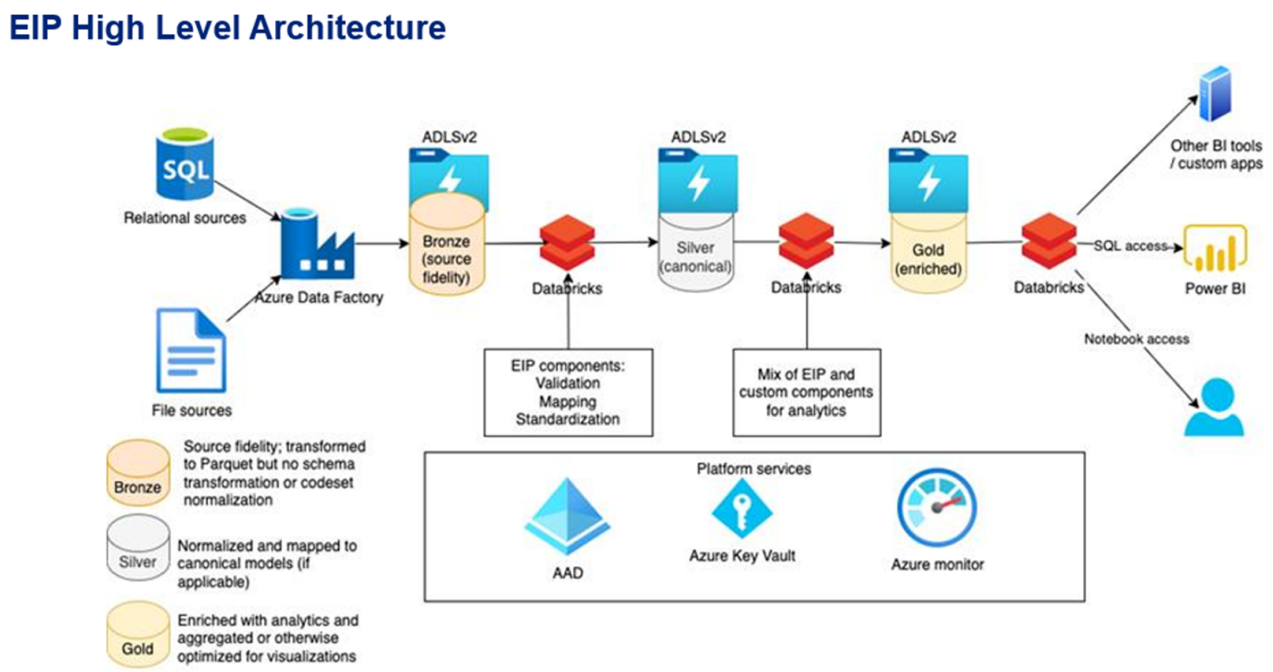
Crimson AI (Formerly EIS) Production Support Operation Training Plan - Onboard Production Team

This space is created to outline the Crimson AI Training Plan for the Production Support Team. For swift onboarding the plan comprises of three parts - The Product and Process Trainings, Technical Trainings and the Access needed to effectively perform operations.

Process



EIP Process and Product Training Plan

CCC- Crimson AI Module

Day	Training	Duration	Description	Training Expectation	Trainer / SME	Initial Completed Date	Training Materials
Day 0	Request access to tools & technologies		Access to different Environments, Tools, workflows & applications.	Guidance to create access requests to different Application, workflows & tools. (Covered in detail in '**Access** ' section)	Self Identification of Required tools and Access As identified per module and area during trainings **For Prod Support**: Create a wiki page and document all the necessary tools and access required based on the inputs gathered from the different trainings.		Access section of this page

Day 1	Introduction to EIP (Crimson AI)	1 Hr.	Application overview, Clients ,teams	• Overview of Power BI based UI • Different metrics & usage • Dashboards & Reports • List of Clients • Crimson AI teams\Stakeholders	Vijay Vaidyanathan	02 May 2024	Video - Day 1
Day 2	Functional Overview	1 Hr.	Deep dive into metrics	Deep dive into metrics, functionalities & Usage	Physician review - Venkappa, Sirigiri <sirigiri.venkappa@optum.com>;	03 May 2024	Video - Day 2 and Optum Learning Community
Day 3	Functional Overview	1 Hr	Deep dive into metrics	Deep dive into metrics, functionalities & Usage	Hospital Quality Optimization - Venkappa, Sirigiri <sirigiri.venkappa@optum.com>;	10 May 2024	Video - Day 3 and Optum Learning Community
Day 4	Architecture overview	1 Hr.	Brief overview of the 3 different layers	• Crimson AI Current Architectural setup • Data flows from client source to the Application	Dhakshinamoorthy, Soundararajan <soundar@optum.com>	16 May 2024	Video - Day 4
Day 5	Data Acquisition workflow (Bronze)	2Hrs	Data Ingestion layer (Raw Layer), raw data conversion to Parquet file format. Source to Silver Layer.	• What happens in this layer • How raw data transformed to Parquet format • Tarnished and Polished Bronze layer • Parquet file overview. • Different level of monitoring and validation steps • Module wise spec understanding	Rajan, Prabhu <RajanP@Optum.com>	1st Session - 12 Jun 2024 2nd Session Continuation 18 Jun 2024	Part 1: Video - Day 6 Part 2: Video - Day 5
Day 6	Data Enrichment workflow (Silver)	2Hrs	Source for Gold Layer (Load layer), Data normalization with general enrichments (Not domain specific enrichment) and applying CDM model concept (Clinical Data Model/ Common Data Model) using FHIR Standard.	• What happens in this layer • Dataflow and transformation. • Different stages of data flow within this layer • Smith Tool/ FHIR Standard Overview • File/ Schema Overview • Different level of monitoring and validation steps	David Raj, Nalin <RajN@Optum.com>	1st Session - 19 Jun 2024 2nd Session - 20 Jun 2024	Part 1: Video - Day 7 Part 2: Video - Day 8
Day 7	Data Enrichment workflow (Gold)	2 Hrs.	Source for Front End visualization(Internal Layer), data enriched with analytics (Facts and Dimension)	• What happens in this layer • Dataflow and transformation. • Different stages of data flow within this layer • File/ Schema Overview • Different level of monitoring and validation steps * ETL Deployment & ADF pipeline	Raja, Avinash	26 Jun 2024	Video - Day 10
Day 8	User Management - Authentication & Authorization	30 mins	Covers user creation, password change & other requests	Areas expected to be covered: • How to Bulk create users • Ticketing and process flow to create user requests, Teams & POCs • Authentication (OHID process) • Authorization tools – Guardian process walk through • Guardian – Roles and provisioning • Any adhoc topic surrounding the User Management area	Vikasitha, Kokkonda Kumar Singh, Shyam	25 Jun 2024	Video - Day 9
Day 9	Data Refresh & Validation	1 Hr.	Deep dive into ETL process & Data validation	• Demo on Data refresh • Demo on DVP tool • Deep dive on Validation measures • How to Trouble shoot in different layers • Best practices	Krishnamoorthy, Sunitha	08 Jul 2024	Video - Day 11

Day 10	Monitoring & ITSM process	1 Hr.	Infrastructure & Performance monitoring	• Overall infrastructure monitoring. Metrics, Alerts & Monitoring tool (if any) • Incident POCs • TAT & Priority categorization	PM, Rakesh <rakesh.pm@optum.com> Joseph, Rajiv	10 Jul 2024	Video - Day 12
Day 11	Tools & Technologies	1 Hr.	Technical requirement for the product and Guidance on installation & learning	• List of technologies • List of tools (DAP, Smith etc) • Installation support • Learning tools (Covered in detail in '**Tools & technologies**' section)	Aravindhan, Balamurugan R <aravindb@optum.com> - Andersen, Wyatt <wyatt_andersen@optum.com> Yerramothu, Krishnaveni		TBD
Day 12	Crimson AI Processes	1 Hr.	Different processes & ticketing tools	• Client support ticketing tool (Salesforce) • Other ticketing tools (Rally , AHA etc) • SLA \ Severities\Priorities • New ticket creation process (Mandatory vs optional fields) • Best practices	Diamond, Greg<Greg.Diamond@optum360.com> Erford, Tristan <Tristan.Erford@optum360.com>		TBD
Day 13	Process Shadowing	2Hrs	Covers day to day activities of DAS & Implementation teams	Paired with DAS & Implementation team to understand the processes involved in Crimson AI day-to-day activities.	David Raj, Nalin / Rajan, Prabhu		TBD
Day 14	Process Shadowing	2Hrs	Covers day to day activities of Data refresh and pipeline troubleshooting	Paired with Data pipeline team to understand the processes involved in Crimson AI day-to-day activities.			TBD
Day 15	Overview of Azure /Databricks	1 Hr	Basic overview of how to access data/files for support and dataload troubleshooting	Ability to query data and find file extracts for clients	Praveen Ramachandran		To be scheduled when Day 1-10 are completed.

SPC - Crimson AI Module

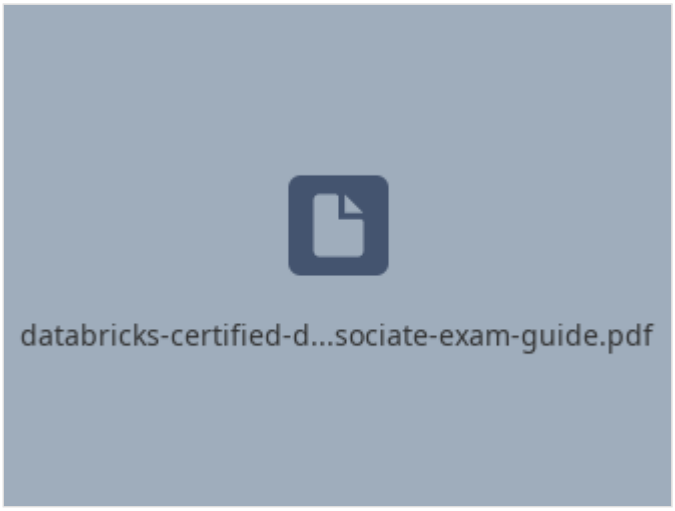
Day	Training	Duration	Description	Training Expectation	Training Materials
Day 1	Surgical Capacity Functional Overview		Deep dive into metrics	Deep dive into metrics, functionalities & Usage	Surgical Capacity
Day 2	Surgical Cost Functional Overview		Deep dive into metrics	Deep dive into metrics, functionalities & Usage	Surgical Cost

Technical

Technical Competency Required:

Technologies	Description	Concepts to Learn	Why in Crimson AI	Reference \Course
Hadoop	An Open Source software utility used for processing massive amounts of data and computation.	Overview of Hadoop architecture. (At fundamental / beginner level) (Like) 1. What is distributed processing system? 2. How large size of data is stored and computed in distributed processing in Hadoop? 3. What is master and Slave node concept? 4. Basic understanding of what is MAP REDUCE concept. 5. Basic overview of Hadoop versioning (1.0 and YARN)	Crimson AI uses Databricks for computing. Basic architectural understanding of Hadoop and Spark will help us to understand the evolution of big data processing and lays the foundation for better understanding of Databricks.	Self-learning: Sample Reference: Great Learning URL: https://olympus.mygreatlearning.com/courses/12888

Spark	Similar to Hadoop, an Open source distributed processing system for big data workloads. However, spark utilizes in-memory caching and optimized query execution. Faster than Hadoop in terms of performance.	Overview of Spark architecture (At fundamental / beginner level). 1. Basic understanding of architectural difference between Hadoop and Spark in terms of processing and computation. 2. Overview of advantages of Spark over Hadoop. 3. Various Spark features like multiple language support, reusability, performance by using in memory caching.	Crimson AI uses Databricks for computing. Basic architectural understanding of Hadoop and Spark will help us to understand the evolution of big data processing and lays the foundation for better understanding of Databricks.	Self-learning: Sample Reference: Great Learning URL: https://www.mygreatlearning.com/academy/learn-for-free/courses/spark-basics Book: Spark : The definitive Guide
Azure	Azure is Microsoft's public cloud computing platform, pay as you use model that provides wide variety of services via internet.	AZ- 900 - This provides basic understanding of what is cloud computing and what are the different services provided in Azure. Overview of how the services and resources are governed and managed in Azure. 1. Basic cloud concepts 2. Overview of Azure compute and networking services. 3. Overview of Azure management and governance compliance DP -900 This offers core data concepts like different data types, different data file formats, different types of data workload, and data services or resources available in Azure. 1. Core data concepts like data format, different data workloads (OLTP, OLAP), file storage. 2. Relational data concepts. 3. Non- relational data concepts. 4. Data Analytics in Azure. DP-203 - This offers core knowledge for data engineer for building, transforming, orchestrating and visualizing an end-to-end data workload using Azure services. DP 203 provides an overview for the following concepts. 1. Roles and responsibility of Data Engineer. 2. Azure synapse analytics. 3. Azure Databricks	Crimson AI product is fully integrated in Azure public cloud platform. AZ - 900 will provide fundamental understanding for all the beginners who are new to Cloud/Azure. DP- 900 & DP-203 will help the Data Engineers/Analyst to understand different types of Data, Data file formats (Parquet file format is used in Crimson AI), different workloads (Analytical workload is used in Crimson AI), different methods of visualization (Power BI is used in Crimson AI for visualization).	Self-paced/ Instructor led learning Course available in: Microsoft URL: https://learn.microsoft.com/

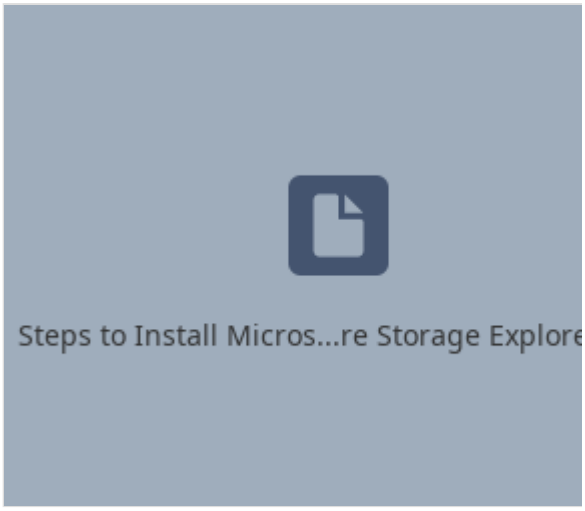
Data bricks	Databricks is an Organization (not an Open source platform) founded by creators of Apache Spark. It provides cloud based platform for big data processing.	Databricks Certified Data Engineer Associate This course offers an individual's ability to use and understand the Databricks Lakehouse platform for Data engineering tasks. 1. What is Databricks? 2. Overview of Databricks Lakehouse platform 3. Databricks architecture and its capability. 4. What is Delta lake and data management with Delta Lake? (Refer reference for more details)	Databricks is used as a computation framework in Crimson AI. Databrick data engineer associate certification will help us to understand, communicate and carry out our day-to-day activities.	Self-learning: Course available in : Databricks Reference: refer to the Document guide attached. 
Power BI	Microsoft visualization tool	Power BI Fundamentals Overview of OLAP (Facts and Dimensions) Overview of most widely used charts available in Power BI Basic understanding of how a chart is created in PowerBI Understanding of Metric navigations and filters	Power BI is used as an visualization tool in Crimson AI. Understanding different charts and different features in PowerBI will help us to drill down and understand the data and different metrics.	Self learning from Microsoft. Refer: https://learn.microsoft.com/en-us/training/paths/get-started-power-bi/ https://learn.microsoft.com/en-us/training/modules/explore-fundamentals-data-visualization/
Scala	Scala refers to scalability, it is a Multi paradigm programming language that supports both functional and Object oriented programming language.	Basic fundamentals of Scala like Datatypes, variables, operators, functions, classes and objects, arrays, collections, loop statements	Predominant work in Crimson AI Databricks is carried out by Scala language. Understanding the Scala code will help the ProdOps to understand the logic written by DT and Engineers (If any)	Book to refer: https://www.oreilly.com/library/view/programming-scala-2nd/9781491950135/ https://docs.scala-lang.org/tour/tour-of-scala.html Wiki page: EIP Spark / Scala, etc. Learning Resources
FHIR & it's Data Types	A common data model that solves the interoperability defines how healthcare data can be exchanged between different computers.	Basic understanding of the following. 1. What is FHIR? 2. Different versioning of FHIR? 3. Basic understanding of different healthcare resources available in FHIR? 4. Detailed overview of FHIR Data types used in Crimson AI.	Crimson AI product workflow is comprised of three layers: Bronze, Silver and Gold. The basic structure of the data that is stored in Silver layers adopts many concepts and structures from FHIR.	Crimson AI Engineering & Self learning from https://www.hl7.org/fhir/overview.html

Access

List of Tools and Applications:

We are still in the process of identifying the mechanism to gain access to the following. Would update this space as more information is available

Line Item	Type	Name	Business Case	URL	Request Via (Secure /Email /Ticket)	Environment (all/ Only prod)	Group Name/ Role Name	Existing document on how to apply if any

1	Application	EIP (Prod /Preprod) -(Front End Application) Authentication & Authorization for Physician Review, Hospital Utilization, Power BI Additional reporting	Validation and replication of the issues.	Stage: https://ei.optum.com/stage?fssocode=uhg-ms-id Prod: https://ei.optum.com/?fssocode=uhg-ms-id	Service Now	Prod		Follow the steps from the following wiki page and refer to the steps mentioned und **User Request"** section. https://confluence.optum.com/display/EIPCOMM/Internal+User+Provisioning+Requ While raising the request in Service now, please fill the following columns with the corresponding details. **Provide a Description:** Email: xxx@optum.com Employee ID: MSID Environment: Prod **Provide Client Information**: Provide the list of clients for which we need access. Example: Willis knighton & St. Bernard. **Provide a product**: Provide module information for which we need to get access. Example: Physician Review, Hospital Quality Optimization, Hospital Utilization Opti Spotlight, Advanced Reporting. **Note**: Tenant Admin is not meant for all. At present, raise application access using now for the clients moved to Production support.
2	Azure	Subscription(Prod /Preprod) for all EIP resource groups like blobs /data lake, keys, VMs, Azure Databricks, ADF pipelines, DB etc	Troubleshooting, to enable us to work on our day-to-day support tickets. To view all the data in different layers like Raw (Bronze), Transformed data (Silver) and aggregated data (Gold).	Crimson AI Prod Access and PIM process - EIP Core Delivery (EIPCORE) - Optum ATLSO Confluence RBAC access documentation	Secure. **Then follow** . Crimson AI Prod Access and PIM process - EIP Core Delivery (EIPCORE) - Optum ATLSO Confluence to activate client based resource group.	Prod	**AZU_7fbc05ce_2e95_4703_ab85_4491e800cc5c_prod_pos_global_JIT**	**To request access:** 1. Go to Secure 2. "Windows" (under "Request New Access") 3. Select "Azure" and "UHG Azure" under "Platform" 4. Select your Azure ID/ Cloud ID (Primary id) 5. Search for the group # **AZU_7fbc05ce_2e95_4703_ab85_4491e800cc5c_ obal_JIT** 6. Click "Next" and submit your request!
3	Tools	Microsoft Azure storage explorer	Useful tool to access the Azure blob storage or ADLS without login into Azure portal.		AppStore			Step by step process on how to install Microsoft Storage explorer. 

4	Application	GitHub	To view all the code repository	Some of the GitHub link related to Crimson AI. Smith Tool (FHIR Conversion logic) : smith-tools/src/main/scala/com/optum/insights/smith/fhir/Encounter.scala at main · uhg-internal/smith-tools (github.com) Crimson AI Infra high level diagram - eip-docs/draw.io/HCC_Prod_Architecture.drawio at main · OptumInsight-Analytics/eip-docs (github.com) Tenant wise Role based Secure Access list: RBAC access documentation	Secure	Prod	"AZU_GHEC_USERS" & "AZU_GHEC_OI_ANALYTICS_USER"	GitHubAccess.pdf
5	Data Extraction	DAP (Data Acquisition Platform)		https://dapdocs.optum.com/capabilities/onboarding/onboarding.html				We need only GitHub access for viewing all the DAP related configurations. Refer to the URL# https://dapdocs.optum.com/capabilities/onboarding/onboarding. all the instructions related to DAP process.
6	Data Validation	DVP (Data Validation Platform)		https://CrimsonAI-prod.qualityportal.optum.com/	Service Now https://optum.service-now.com/itss2/			Follow the instruction as mentioned in the following wiki page for DVP access Run Book For DVP URL Access

7	CMT	Common Mapping Solution		• CMT Stage - https://stg.cmt.data-enablement.optum.com/ • CMT Prod - https://cmt.data-enablement.optum.com/	Secure DFE_CMT_EIP_DATA_ANALYST_PROD			Request via Secure to access CMT.
8	Ticketing Tool	Salesforce						
9	Ticketing Tool	Optum AHA						This involves payment and not sure if it is necessary for all team members.
10	Ticketing Tool	Rally						
11	Ticketing Tool	Service Now						
12	Ticketing Tool	RITM						

13	Monitoring Tool	Splunk	To view all the Data related log. Log Repository	https://splunk.o360.cloud/ Index: • oi_eip_nonprod • oi_eip_prod				No special permission or ticket is required. By default, we should able to access th
14	Monitoring Tool	VictorOps						
15	Monitoring Tool	SignalFX	To view Infra related monitoring metrics & Dashboards	Dashboards - Crimson AI - Enterprise Intelligence Suites - formerly EIP - Enterprise Intelligence Platform (signalfx.com)				No special permission or ticket is required. By default, we should able to access th
16	Monitoring Tool	Splunk Synthetics						
17	Access Tool	Guardian	User management	https://developer.linkhealth.com/apps/security/EIS	Secure & RITM	Prod	**Secure Application** Link Platform (EIS) **Environment** Test (For Staging) Prod (For Production) **Roles** EIS User **RITM**	

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